

An Overview of the Contemporary GSM System in Digital World

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Abstract

This undertaking is geared toward the layout and implementation of the continued studies on the M2M wi-fi communicate of diverse machines and gadgets in cellular networks is a fast-growing business and application location in the enterprise, maintenance enterprise, customer service, safety, and banking areas. This assignment presents the layout and implementation of a far flung manipulate system by using the GSM cell verbal exchange network. The design integrates the tool to be managed, the microcontroller, and the GSM module so that it is able to be used for a huge variety of programs. On this project, we aimed to switch off/on the home equipment via sending the message.

Keyword: Encryption, GSM, GPRS, microprocessors, Wi-Fi

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INTRODUCTION

The concept of a clever domestic is an emerging trouble to the modern-day era-established society. Far off manipulate technologies are extensively used to control household electronic appliances without on foot up to them. Controlling family appliances via the laptop also can be a possible answer. But, it can't satisfy the current demand that's to manipulate them from remote places [1]. The benefit of cellular communications like GSM era is a ability answer for such remote controlling activities. Domestic automation structures are an end result of an try and enhance the requirements of living around and in the residence and were made possible with the aid of the breakthrough in the area of incorporated circuits and microprocessors. These systems provide the consumers with increased security and safety, economic benefit through energy control, and convenience by giving them control over every piece of domestic electrical equipment in the house [2].

OVERVIEW

The project is based on M2M technique. M2M allows a machine or device to transmit or receive its data remote area over a network.

These projects demonstrate the switch off/on of the home appliances by the means of GSM cellular network via message [3]. The project uses GSM module and microcontroller and relays for controlling the appliance. The project is having four relays for connecting the devices of rating 220V/10A such as fan or bulb for being switch off/on via SMS [4]. as shown in Figure 1.

EXPLANATION OF PROBLEM

Normally it's miles found that once the user is far from the appliances he/she is unable to function the home equipment or it very commonly happens that we forgot to switch off the home equipment because of which useless power voltage is ate up [5]. And the greater cost has to pay for the labor for working the home equipment manually, as the problem diagram shown in Figure 2.

Improvement in Contemporary GSM

Controlling of the home equipment from the far flung vicinity through SMS, DTMF, or through call make it an excellent solution for plenty issues including reducing the price (e.g. The value of the labors that manipulate or take a look at the system might be reduced by controlling and checking the

machine remotely), and reducing the time of monitoring and controlling the system (mainly when the device is very some distance, the time to get to the gadget may be very long in comparison to a single click or sending a simple text message). Machines and devices may be vending machines, elevators, pumps, meters, traffic lighting fixtures, and presentations. As shown in Figure 3.

WORKING of GSM

M2M enables the glide of information among machines and machines and in the end Machines and those. Regardless of the type of gadget or information, facts commonly flows from a machine over a community, and then through a gateway to a gadget where it is able to be processed and analyzed. GSM I/O Relay Board is a superb development board for cellular and remote monitoring programs [6].

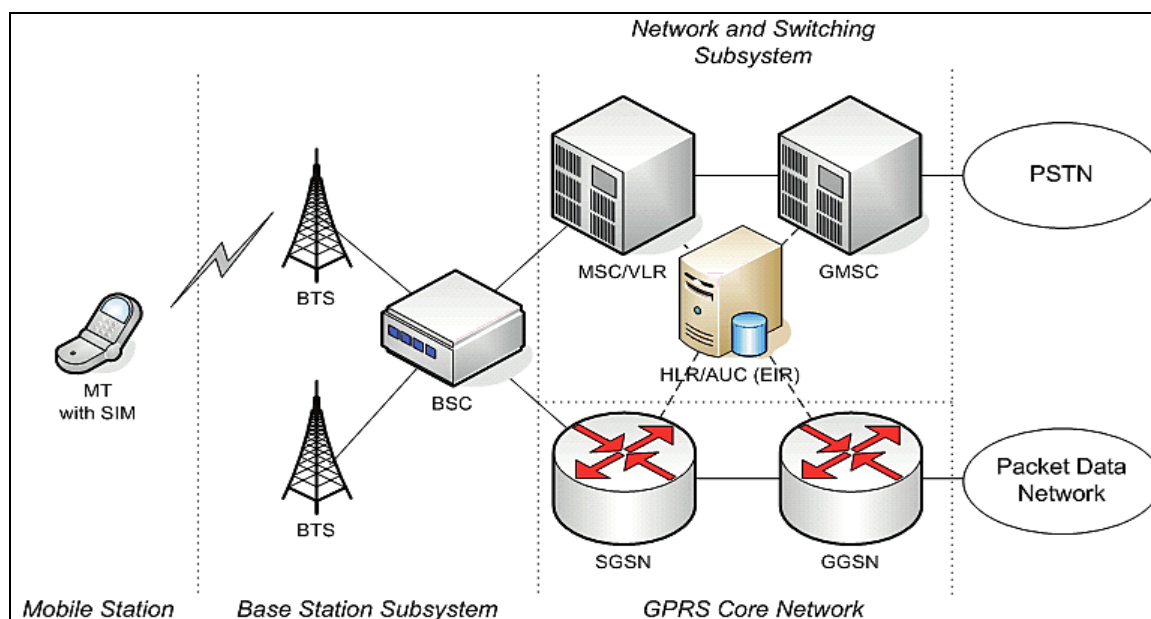


Fig. 1: An Overview of GSM System.

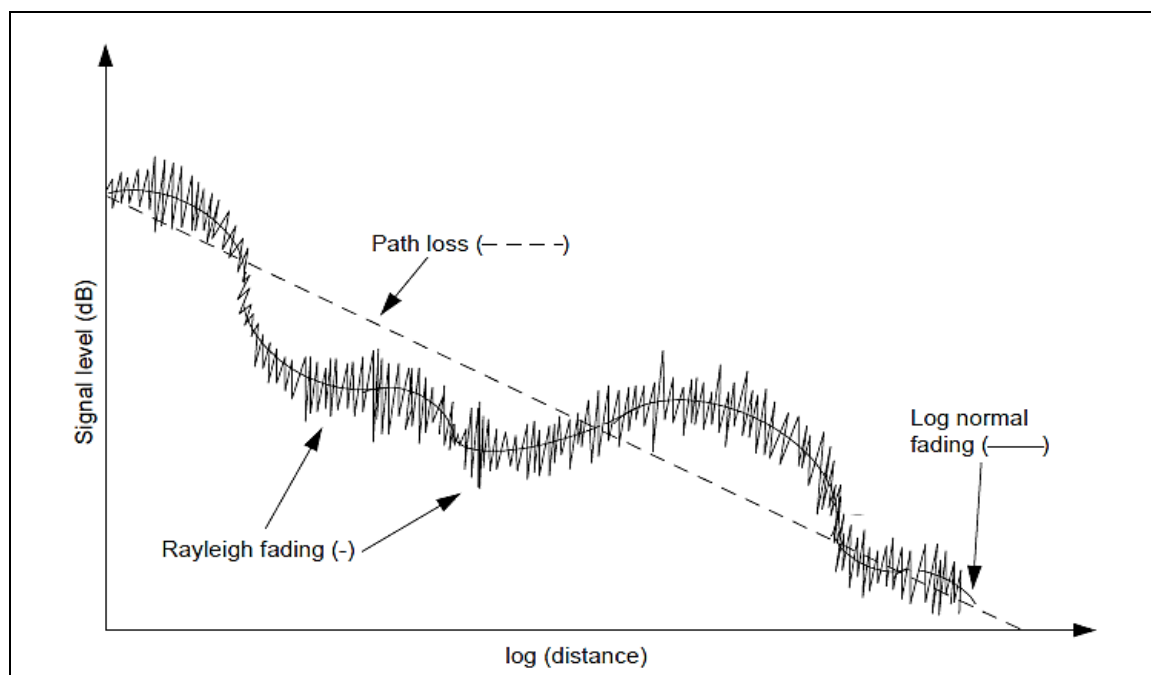


Fig. 2: Problems Faced in GSM.

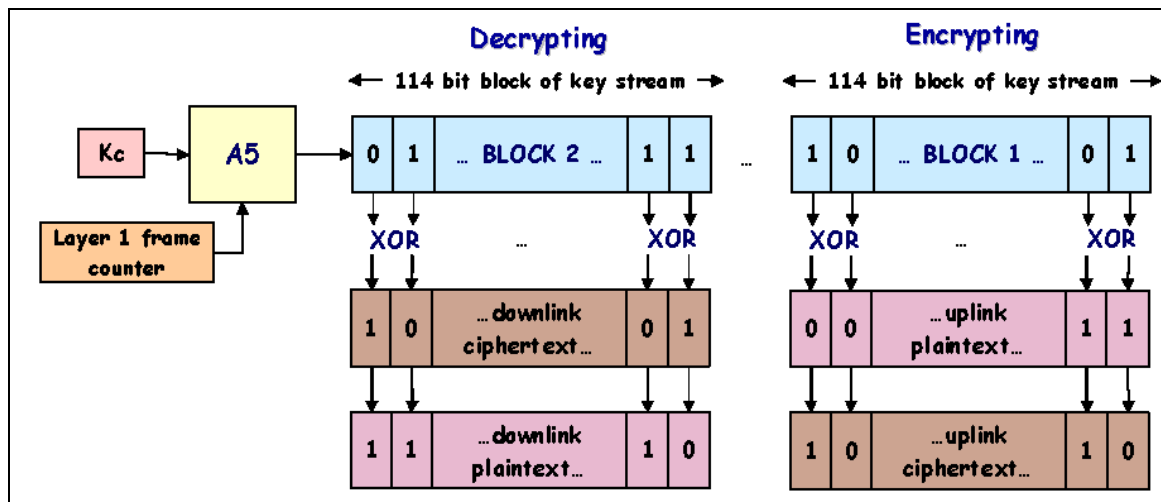


Fig. 3: A Contemporary GSM.

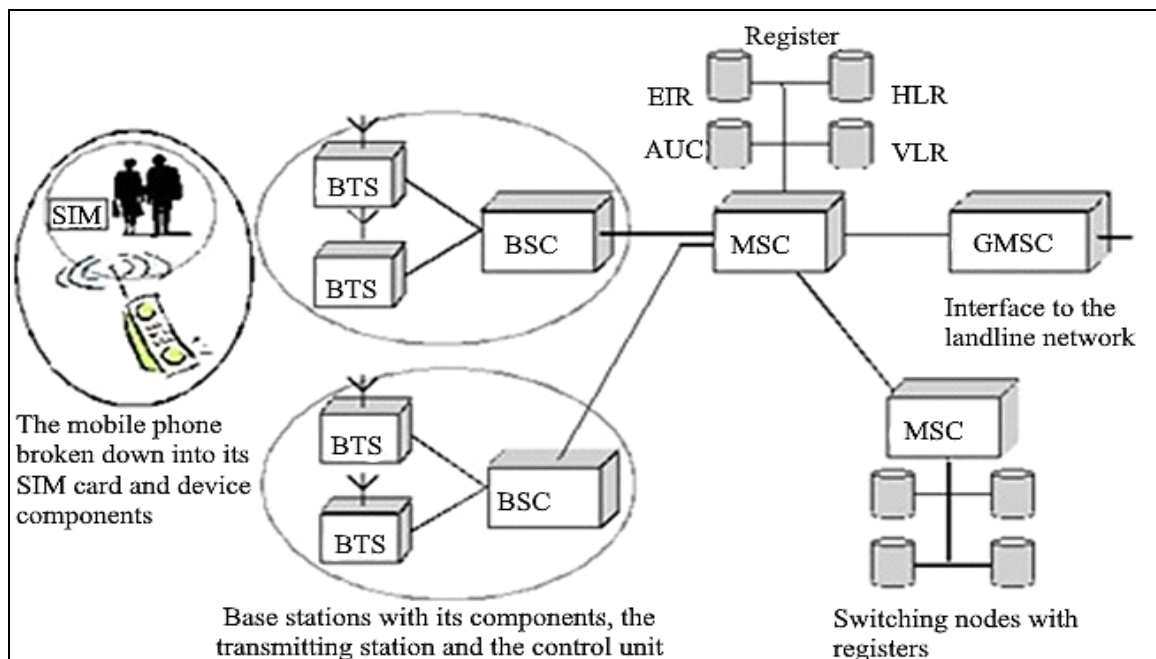


Fig. 4: Working of GSM.

It is built with Triband GSM/GPRS engine. With it, you could easily display and control your own home from anywhere there may be cell phone coverage along with your GSM cell account working is shown in Figure 2. Insert any GSM SIM card and begin gaining access to the board remotely over the cellular community thru SMS. When the sim is inserted inside the GSM module firstly it success for the most powerful frequency within the cellular and use it for the conversation. As the message is mailed from the far flung region it first off gets by means of the module and transfers to the microcontroller thru UART port [7] as shown in Figure 4.

Various Aspects of GSM

The controller saves the command and decode it and examine it with the saved program and determine what operation is to be accomplished if the acquired message if no one of the command saved in the microcontroller then it certainly deletes it in any other case it triggers the port according to the command and a relay driving force IC is used to pressure the relay which offers enough quantity of current to power it [8]. This IC is the mixture of the six BJT hence the linked equipment from the relay is transfer off or on. shown in Figure 5.

Architecture of GSM System

GSM I/O Relay Board is a great development board for cellular and remote monitoring applications [9]. It is built with Trainband GSM/GPRS engine. With it, you can easily monitor and control your home from anywhere there is cell phone coverage with your GSM cellular account. Insert any GSM SIM card and start accessing the board remotely over the cellular network via SMS/GPRS. Since it is built with powerful TCP/IP Stack, you can communicate with this board from anywhere in

the world through internet also [10]. Designing an M2M system for monitoring and controlling machine and devices in remote locations can be done through a variety of communications options such as wireless LAN technology, dial-up modems, private radio networks, satellite communication, and cellular network. Out of these options, the cellular network is getting so much attention these days for transporting M2M as given in Figure 6 data, particularly the most widely adopted GSM (global system for mobile communication) [11].

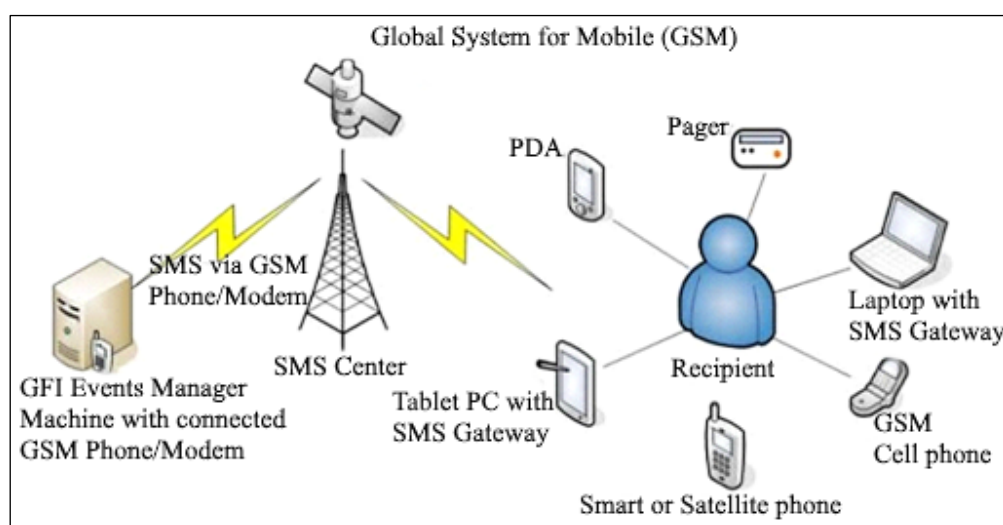


Fig. 5: A GSM Diagram for Mobile Communication.

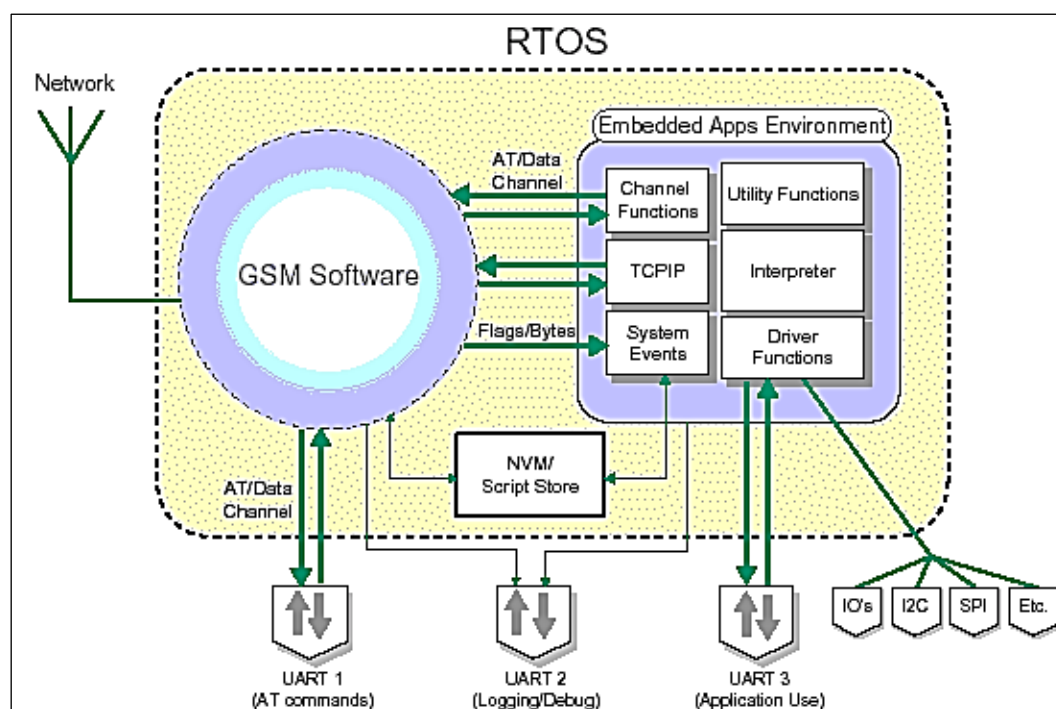


Fig. 6: Architectural Behaviour of GSM System.

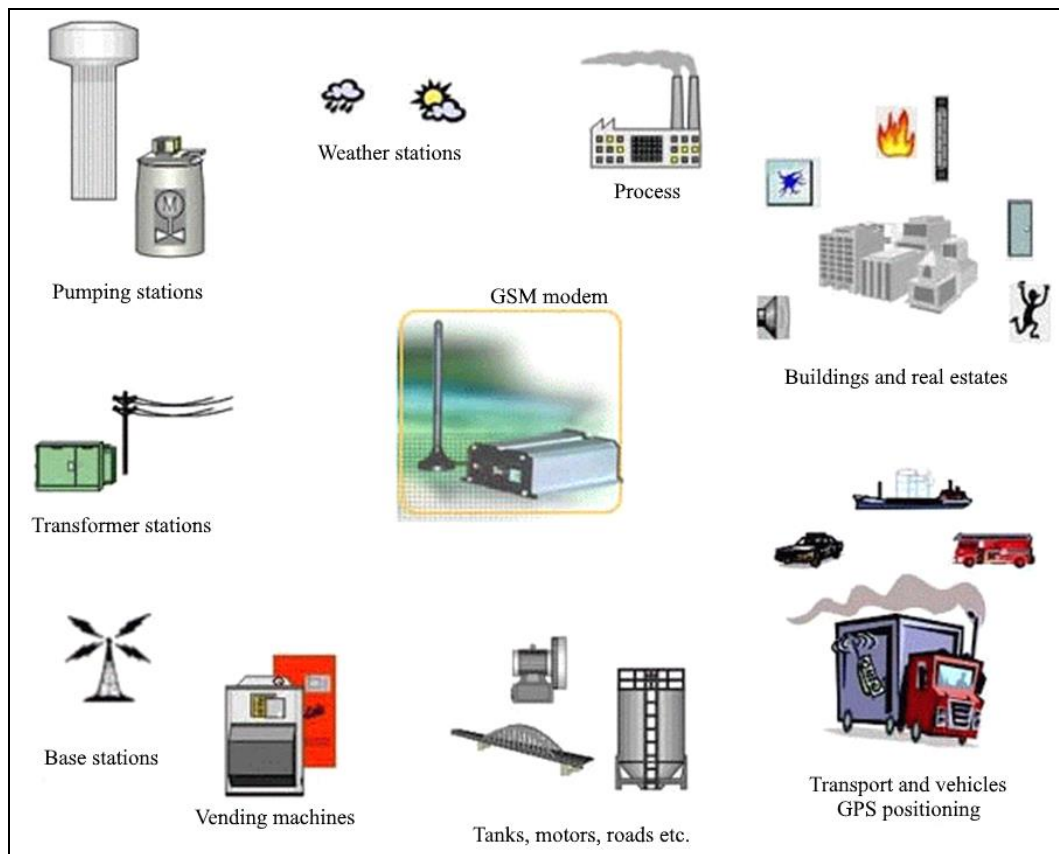


Fig. 7: Various Applications of GSM.

Applications of GSM

Various applications are shown in Figure 7.

- Weather stations
- Transformers stations
- GSM MODEM
- Base stations
- Pumping stations
- Building and real estates
- Industries etc.

CONCLUSION

The device gives a more green and less doable manner of using in the various stations like base, pumping, climate and so forth. The prototype of the gadget provides numerous blessings over traditional method of taking attendance. The prototype advanced is compact and mild weight and may run on strength adapter or battery energy. Similarly, it's miles very transportable and may be moved from one point to the alternative. The gadget can be applied not just in the colleges but additionally within the running places. Therefore, making use of GSM in diverse contemporaries overtaking system i.e. CDMA the usage of contemporary method.

REFERENCES

1. M. Stephen, E. S. Johnson and R. W. John, (2012). RFID Security Technology and Applications, Cambridge University Press.
2. K. Langendoen, A. Baggio, and O. Visser, Murphy loves potatoes: experiences from a pilot sensor network deployment in precision agriculture, in Proc. of 20th International Parallel and Distributed Processing Symposium (IPDPS 06), Rhodes Island, Greece, 2006
3. Yanbo Zhao; Zhaohui Ye; A low cost GSM/GPRS based wireless home security system IEEE Transactions on Consumer Electronics, Issue Date: May 2008, Volume: 54, Issue:2, On page(s): 567.
4. Ying-Wen Bai; Chi-Huang Hung; Remote power ON/OFF control and current measurement for home electric outlets based on a low-power embedded board and zigbee communication. Consumer Electronics, 2008. ISCE 2008. IEEE International Symposium on; Issue Date: 14–16 April 2008 On page(s):

5. Deepa B Chavan, Abdul Rahim Makandar, Automatic Vehicle Speed Reduction System Using Rf Technology, International Journal of Engineering Research and Applications, Vol.4, No.4, 2014.
6. Fleischer, P.B.; Nelson, A.Y.; Sowah, R.A.; Bremang, A. Design and development of GPS/GSM based vehicle tracking and alert system for commercial inter-city buses," Adaptive Science & Technology (ICAST), 2012 IEEE 4th International Conference on, vol., no., pp.1,6, 25–27 Oct. 2012
7. Theodore S. Rappaport (2008), Wireless Communication, Prentice Hall PTR.
8. Salas Jose, Namitha Mathew, X. Mary, ARM 7 Based Accident Alert and Vehicle Tracking System, International Journal of Innovative Technology and Exploring Engineering, Vol-2, Issue-4, 2013
9. U. J. Kameswari, M. Satwik, A. Lokesh, and G. V. Reddy, A design model for automatic vehicle speed controller, International Journal of Computer Applications (09758887) Volume, 2011
10. K. Nandaniya, V. Choksi, A. Patel, and M. Potdar, Automatic accident alert and safety system using embedded GSM interface, International Journal of Computer Applications, vol. 85, no. 6, 2014.
11. N.H. Sane, D.S. Patil, S.D. Thakare, and A.V. Rokade, Real Time Vehicle Accident Detection and Tracking using GPS and GSM, International Journal on Recent and Innovation Trends in Computing and Communication.

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